Results: toy falsifier for the θ_2 Coppersmith ceiling on S_3

Date. 2026-10-23. **Protocol.** `PROTOCOL.md`, frozen and committed (`c785a0e23`) before the run. **Run.** `reports/run-20261023T2122Z/` (`run.md`, `run.json`, `trials.jsonl`, `recovery.svg`, `report.pdf`). **Instrument.** `examples/coppersmith_s3_theta2_falsifier.rs`. **Question.** §7 item 1 of `RESEARCH_LATTICE_ECDLP_ROUTES.md`: does a Coppersmith-type lattice on S_3 recover a planted two-summand decomposition with $x(P_i) \le p^{\beta}$ beyond $\beta = 1/4$?

Verdict (decision rule of the protocol, §1 and §5)

Closed at measured level, for $\ell \le 3$. Recovery was 0 of 10 in every cell with $\beta \ge 0.26$ (60 cells, 600 trials), in fact in every cell with $\beta \ge 0.20$ (96 cells, 960 trials), at all four sizes and all three multiplicities, and every positive control passed. The refutation criterion (≥ 5/10 in any cell with $\beta > 1/4$) was not met in any cell. The §6 ceiling of the note stands at the measured level, and its finite-ℓ rows are confirmed by the Howgrave-Graham column (below).

cell family	result	positive control (same instrument)	verdict wording
all $\beta \ge 0.26$, 4 sizes × $\ell \in \{1,2,3\}$	0/10 in 60 of 60 cells	passed in every (`bits`, ℓ) row that has one (see below)	no recovery above the stop line
all $\beta > 1/4$	max recovered 0/10	—	refutation criterion not met
negative control, $\beta = 0.30$	0 genuine pairs in 120 trials	—	no false positives on random targets

Positive controls (smallest grid β with $B \ge 4$, provided $\beta \le \beta_{det}(\ell) + 0.02$): 24-bit $\beta = 0.10$: $\ell = 2$ 10/10, $\ell = 3$ 10/10; 32-bit $\beta = 0.08$: $\ell = 2$ 10/10, $\ell = 3$ 10/10; 40-bit $\beta = 0.06$: $\ell = 1, 2, 3$ all 10/10; 56-bit $\beta = 0.04$: $\ell = 1, 2, 3$ all 10/10. Rows (24, $\ell = 1$) and (32, $\ell = 1$) have no positive control under the protocol's clause (their $\beta_{det} = 0.056$ gives $B < 4$) and are excluded from the verdict; they are reported in full and behave like the others.

Where recovery actually stops, against the derivation

ℓ	dim	derived $\beta_{det}(\ell)$	HG holds 10/10 up to (24 / 32 / 40 / 56 bits)	recovered ≥ 5/10 up to	last nonzero recovery
1	9	1/18 = 0.056	0.06 / 0.06 / 0.06 / 0.06	0.12 / 0.08 / 0.08 / 0.08	0.12 / 0.12 / 0.12 / 0.08
2	25	1/10 = 0.100	0.08 / 0.08 / 0.10 / 0.10	0.14 / 0.12 / 0.12 / 0.12	0.18 / 0.14 / 0.14 / 0.12
3	49	5/42 = 0.119	0.10 / 0.10 / 0.10 / 0.10	0.14 / 0.14 / 0.14 / 0.12	0.16 / 0.14 / 0.14 / 0.14

Reading (classes per the controlled-experiment skill):

- **Howgrave-Graham (the rigorous sufficient condition) tracks $\beta_{det}(\ell)$ to within the grid step at every size.** Measured: holds through 0.06 / 0.08–0.10 / 0.10 for $\ell = 1$ / 2 / 3, against derived 0.056 / 0.100 / 0.119. This is the determinant condition of §6 of the note, measured rather than taken on faith. Comparative, established by this run on these cells.
- **Heuristic recovery runs 0.02–0.06 above the rigorous bound, and the excess shrinks as the prime grows** ($\ell = 1$: last recovery 0.12 at 24 bits, 0.08 at 56 bits; $\ell = 2$: 0.18 → 0.12; $\ell = 3$: 0.16 → 0.14). LLL finds shorter vectors than the worst-case bound promises, and the margin measured in β is a constant number of bits that is worth less as `bits` grows. Comparative; the trend is in the direction the $p \to \infty$ analysis predicts and is not a claim about the limit.
- **Nothing approaches rho parity.** The stop line $\beta = 1/4$ is 0.07–0.17 above the last recovery of any cell; the $\ell \to \infty$ limit 1/6 ≈ 0.167 was crossed by a single 1/10 cell (24 bits, $\ell = 2$, $\beta = 0.18$) and by nothing at 32 bits or more. Exact count on the recorded cells.
- **Multiplicity buys little.** Going from $\ell = 1$ (dim 9, ≈ 1–7 ms per reduction) to $\ell = 3$ (dim 49, 0.1–7 s) moves the recovery threshold by about 0.04–0.06 in β, as β_{det} predicts (0.056 → 0.119); the asymptote is 0.167. Comparative on the recorded cells.

Effect size in the unit that matters: at 56 bits the largest recovered bound is $B = 105$ ($\ell = 3$, $\beta = 0.12$, 10/10) against the $B = p^{1/4} \approx 1.6 \cdot 10^4$ that rho parity would need and the $B = 539027$ ($\beta = 0.34$) that was tested and failed 0/10.

Scope: what this does not show

- Only the Jochemsz–May basic box lattice of the single polynomial S_3, at $\ell \le 3$, with the repository's LLL (no BKZ, no extended shifts, no symmetric rewriting; Probe C and W10 cover those and found the same threshold). It is a measured ceiling at these multiplicities, not a proof of the $\ell \to \infty$ limit and not a statement about other lattice families.
- Four sizes, one curve each, $N = 10$ per cell: a 0/10 cell bounds the recovery probability by 0.31 (95% one-sided). The verdict rests on 60 such cells agreeing and on the monotone shape of every row, not on any one cell.

- Not an attack, a speedup or a security statement. Under the success criterion this is a boundary measurement of a derivation: it changes no baseline and reaches no leaderboard.

Prior measurements this agrees with

`RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §3.2 (Probe B, 40 bits: "≥ 2 rows vanish" to $\delta = 0.12$–0.14, Howgrave-Graham to 0.10), §3.3 (Probe C, extended and symmetric lattices, same threshold), and `research/lit2026_w10_s5_coppersmith_20261011/` §5.4 (32 bits, $\ell = 1, 2$). This run adds genuine root extraction as the metric (the resultant step, candidates checked against S_3), two further sizes, $\ell = 3$, the sweep past 1/4, and the two control arms, under a frozen protocol.

Checklist (controlled-experiment skill §6)

- ☒ every published sentence classified (heuristic estimate tested; comparative statements marked; exact counts marked)
- ☒ negative-control population stated (random targets, 120 trials)
- ☒ every "nothing found" has a passing positive control on the same path
- ☒ one instrument, identical budgets across arms, seeds recorded
- ☒ resolution arithmetic in the protocol (per-cell count; family false-refutation probability below $4 \cdot 10^{-3}$)

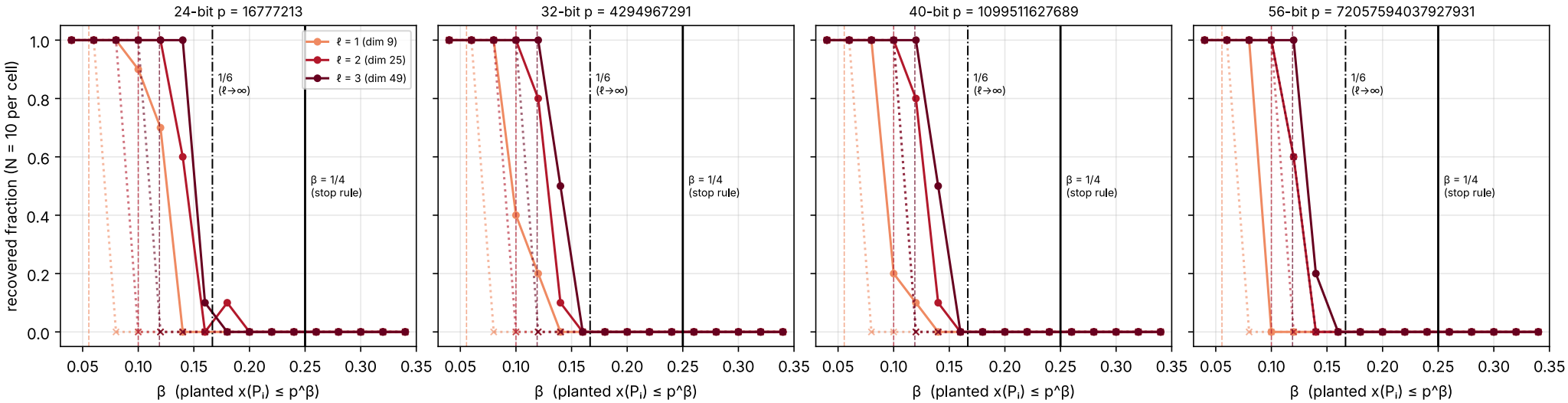


Figure 1: Recovered fraction per cell; the run report has the table